# MATH70110: QRM Coursework 2

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**Abstract**

We respond to Bob's position change to TSLA with a tail-focused risk analysis and forecast. In contrast to the previous coursework, where we focused on empirical, non-parametric and adaptive approaches involving rolling windows and historical simulations, this coursework focuses on providing theoretically stable and interpretable parameter-based risk forecasting based on Extreme Value Theory. We discover that the normal distribution performs fairly well within $0.95 \leq \alpha \leq 0.99$ for unconditional coverage backtesting. However, a quick experiment on extreme levels ($\alpha > 0.99$) reveals that this is where the Generalised Pareto Distribution shines.

## 1 (i) GARCH(1,1) fitting

We fit a standard GARCH(1,1) model with constant conditional mean and standard normal innovations using log-return observations prior to 26 November 2021. The model is as follows.

$$r_t = \mu + \sigma_t Z_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \qquad Z_t \sim N(0,1).$$

The model fitting results are shown in Table 1:

Table 1: Estimated GARCH(1,1) parameters on TSLA training data [Appendix 2]

| Parameter | Estimate | Std. Error |
|---|---|---|
| $\mu$ | 0.1506 | 0.0642 |
| $\alpha_0$ | 0.1390 | 0.0782 |
| $\alpha_1$ | 0.0409 | 0.0133 |
| $\beta_1$ | 0.9482 | 0.0177 |

The estimated parameters on Table 1 show that the conditional mean $\mu$ is small relative to the volatility of the TSLA returns. The GARCH parameters satisfy the positivity conditions, and the persistence $\alpha_1 + \beta_1 = 0.9891$ is close to one, which indicates highly persistent volatility. This is consistent with the well-known behavior of equity returns, where periods of high volatility tend to cluster. The estimated $\alpha_1$ is modest relative to $\beta_1$, implying that recent shocks have a smaller immediate impact than the long-term volatility component.

Observing Figure 1, we see that the ACF of the standardised residuals is flat, and the same is true for the ACF of absolute residuals. This suggests the GARCH(1,1) model has removed most

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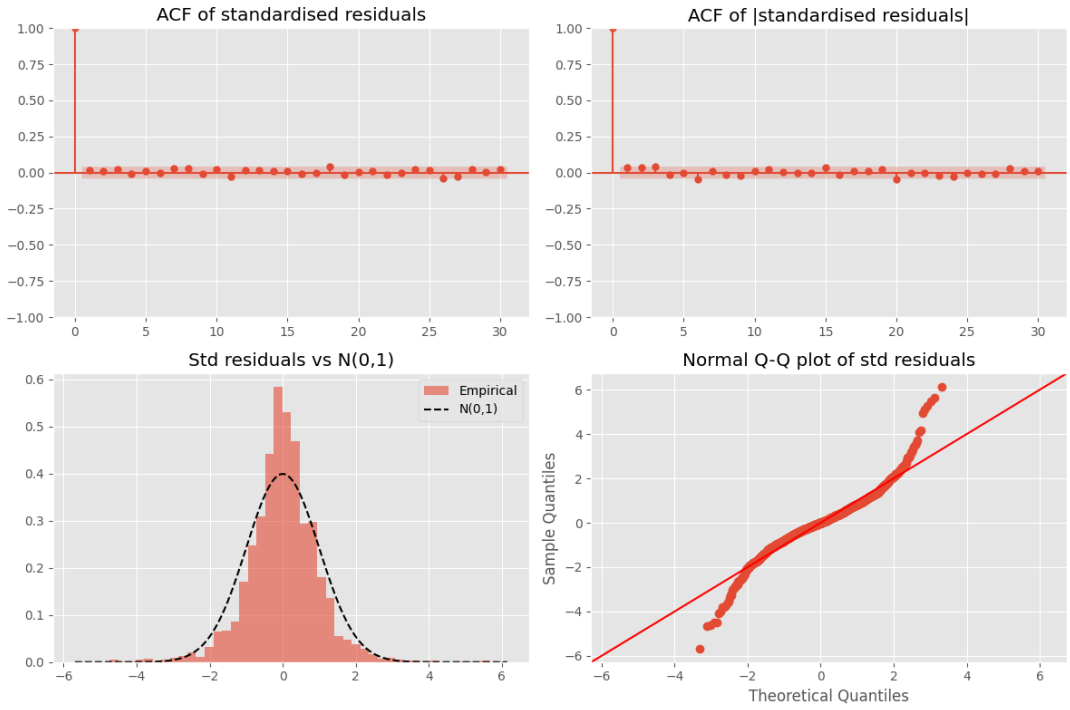



Figure 1: Residual diagnostics for the GARCH(1,1) model: ACF of standardised residuals, ACF of absolute residuals, histogram vs $N(0,1)$, and QQ-plot. [Appendix 2]

of the dependence in both the returns and their volatility. The histogram is more peaked in the centre and has heavier tails than a normal distribution, which matches the high sample kurtosis (around 7.5). The Q-Q plot also shows clear deviations in both tails. Hence, the residuals look roughly i.i.d. in terms of dependence, but they are not well described by a normal distribution because of the heavy tails.

## 2 (ii) Student-t fitting on residuals

To improve the modelling of heavy tails in the GARCH residuals, we fit a normalised Student-$t$ distribution to the standardised residuals $\widehat{Z}_t$ obtained in part (i). If $Y \sim t_\nu$ follows a standard Student-$t$ distribution with $\nu$ degrees of freedom, we define the transformed variable

$$Z = \sqrt{\frac{\nu-2}{\nu}}\, Y,$$

so that $\mathrm{Var}(Z) = 1$. This parameterisation ensures compatibility with the standardised GARCH residuals.

The degrees of freedom parameter $\nu$ is estimated by maximising the log-likelihood

$$\ell(\nu \mid z_1, \ldots, z_n) = \sum_{t=1}^{n} \log f_{t_\nu}\big(z_t; \mathrm{loc} = 0, \mathrm{scale} = \sqrt{(\nu-2)/\nu}\big),$$

under the necessary constraint $\nu > 2$. Using numerical optimisation (L–BFGS-B), we obtain the estimate

$$\widehat{\nu} = 3.800.$$

This relatively low value of $\nu$ indicates substantial tail heaviness, confirming that the Gaussian distribution used in part (i) is insufficient to capture extreme return behaviour.

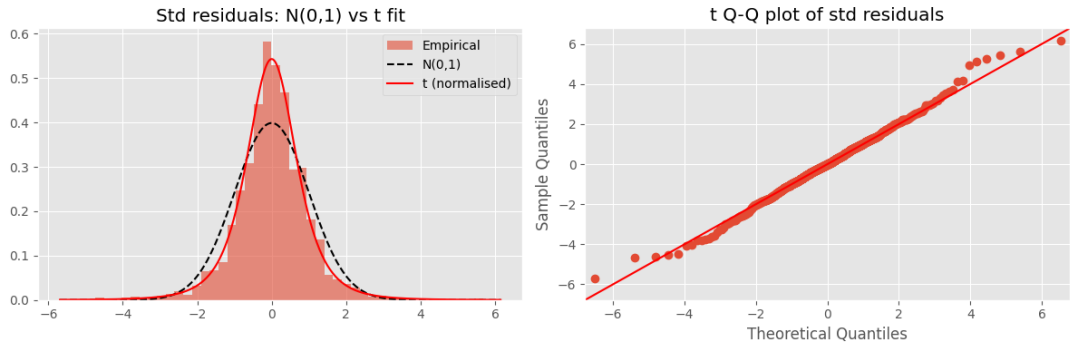



Figure 2: Residual diagnostics for the fitted Student-t model. [Appendix 3]

Figure 2 compares the empirical density of the standardised residuals with both $\mathcal{N}(0,1)$ and the fitted normalised Student-$t$ distribution. The fitted Student-$t$ curve matches the empirical tail behaviour much more closely than the Gaussian benchmark. The associated Q–Q plot also shows a significantly improved alignment in the extreme quantiles, indicating a markedly better tail fit.

Overall, the normalised Student-$t$ distribution provides a substantially improved representation of the standardised residuals and captures the heavy-tailed structure of TSLA returns effectively.

# 3 (iii) GPD fitting on residuals

To model the extreme downside risk of TSLA, we apply conditional Extreme Value Theory (EVT) to the upper tail of the transformed residuals $Z_t = -\widehat{Z}_t$. This allows us to capture large negative return shocks through the tail distribution of $Z_t$.

We first compute the sample mean excess function

$$e_n(u) = \frac{1}{N(u)} \sum_{t:Z_t>u} (Z_t - u),$$

over a grid of high quantiles. A region where $e_n(u)$ appears approximately linear indicates suitability for the GPD approximation.

Inspection of Figure 3 shows two approximately linear regions beginning at around $u \approx 0.80$ and $u \approx 1.72$. We select the earlier value in order to capture the most data. We therefore set the EVT threshold to

$$u_Z = 0.80,$$

producing 357 exceedances out of 2266 observations, which is a reasonable proportion for stable GPD estimation.

Let $Y_j = Z_j - u_Z$ denote the excesses. We invoke the Pickands-Balkema-de-Haan theorem, which states that for a broad class of underlying distributions $F$, the conditional distribution of excesses over a sufficiently high threshold $u$,

$$F_u(y) = P(Z - u \le y \mid Z > u) = \frac{F(u+y) - F(u)}{1 - F(u)}, \qquad y \ge 0,$$

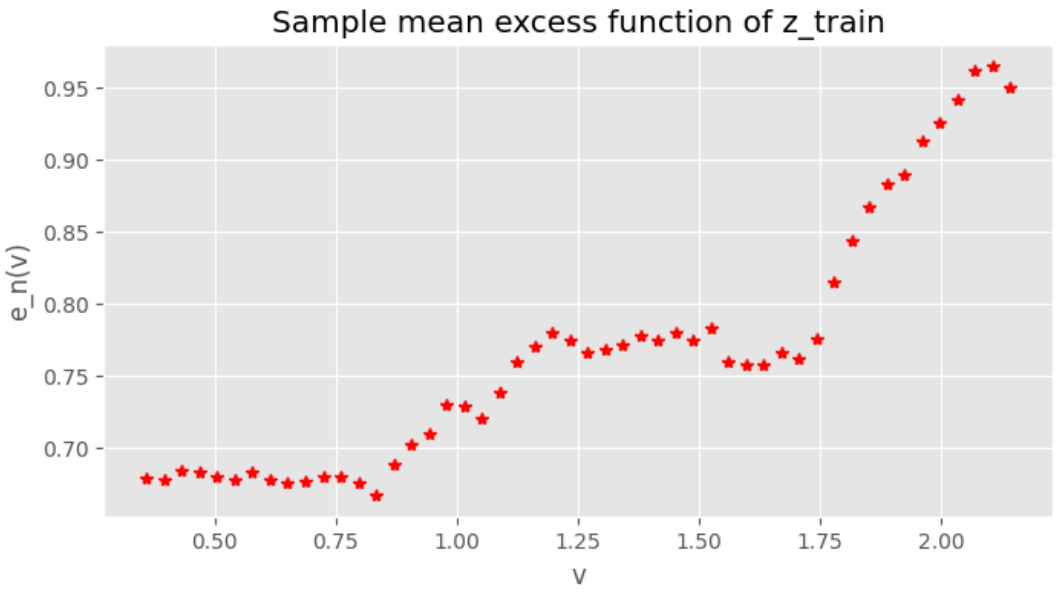



Figure 3: Sample mean excess function of $z_{\text{train}}$. [Appendix 4]

converges to a GPD as $u \to x_F$, where $x_F$ denotes the right endpoint of $F$. That is,

$$F_u(y) \approx G_{\xi,\beta}(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi}, \qquad y > 0,$$

for some shape parameter $\xi$ and scale parameter $\beta > 0$. We fit the distribution via Maximum Likelihood optimization, based on the approach presented by the lecture note utilizing the minimise function in scipy. We initialise $(\xi, \beta)$ at $(0.5, 1.0)$, corresponding to a moderately heavy tail and a scale compatible with the magnitude of the exceedances, and constrain the search to $0.01 \leq \xi \leq 5$ and $0.01 \leq \beta \leq 10$ to enforce the GPD's support conditions and maintain numerical stability while still covering all economically plausible parameter values. We obtain the parameter estimates

$$\widehat{\xi} = 0.138, \qquad \widehat{\beta} = 0.583.$$

The positive but moderate shape parameter $\widehat{\xi} > 0$ confirms that the tail is heavy but not extremely so; this is characteristic of financial return data. The empirical vs. fitted CDF in Figure 4 (plotted on a log scale in $x$) shows good agreement, especially in the upper region of the tail. Minor deviations in the extreme uppermost points are expected given the sparsity of observations.

The EVT analysis supports the presence of heavy-tailed downside risk in TSLA returns. The fitted GPD provides a statistically adequate description of tail exceedances and can be reliably used for subsequent estimation of tail risk measures such as Value-at-Risk (VaR) and Expected Shortfall (ES). Combined with the Student-$t$ residual model, this EVT-based tail modelling forms a robust framework for quantifying extreme return risk for TSLA.

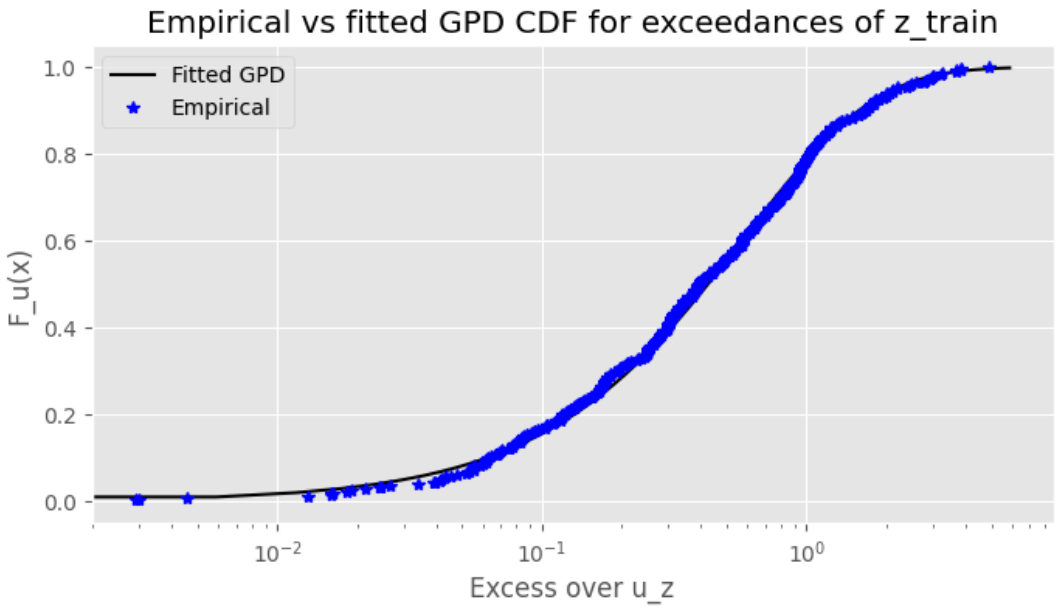



Figure 4: Empirical vs fitted GPD CDF [Appendix 4]

# 4 (iv) VaR/ES forecasting

In this section, we will explain the methodology for forecasting VaR and ES for any confidence level ($\alpha$). We use innovations that follow $N(0,1)$, normalised Student-$t$ with degrees of freedom $\hat{\nu}$, and a GPD tail with parameters $(\hat{\xi}, \hat{\beta})$ beyond the threshold $u$.

First, we need a mathematical model to describe the losses. Here, we model losses $L_t$, $t \in Z$, using a causal and invertible ARMA–GARCH process. Thus,

$$L_{t+1} = \mu_{L,t+1} + \sigma_{t+1} Z_{t+1},$$

where $(Z_t)_{t \in Z} \sim \text{SWN}(0,1)$. However, in practice,

$$\hat{L}_{t+1} = \hat{\mu}_{L,t+1} + \hat{\sigma}_{t+1} Z_{t+1} \tag{1}$$

Since $\bar{L}_{\Delta,t+1} = -r_{t+1}$ and the conditional mean of returns is taken to be constant $\hat{\mu}$, we have

$$\hat{\mu}_{L,t+1} = -\hat{\mu}$$

Moreover, GARCH(1,1) allows us to forecast $\hat{\sigma}^2_{t+1}$ using the approximation,

$$\hat{\sigma}^2_{t+1} = \hat{\alpha}_0 + \hat{\alpha}_1 r_t^2 + \hat{\beta}_1 \hat{\sigma}_t^2,$$

where $\hat{\alpha}_0$, $\hat{\alpha}_1$ and $\hat{\beta}_1$ have already been computed. From (1), the VaR and ES for any $\alpha$ can be obtained as follows:

$$\hat{VaR}_\alpha(L_{t+1}) = \hat{\mu}_{L,t+1} + \hat{\sigma}_{t+1} q_\alpha(Z_{t+1})$$

$$\hat{ES}_\alpha(L_{t+1}) = \hat{\mu}_{L,t+1} + \hat{\sigma}_{t+1} ES_\alpha(Z_{t+1})$$

where $q_\alpha$ and $ES_\alpha$ denote the $\alpha$-quantile and the expected shortfall at level $\alpha$, respectively. These values take different forms according to the distribution assumed by $Z_{t+1}$. If one assumes

$$Z_{t+1} \sim N(0,1),$$

then for any $\alpha \in (0,1)$,

$$q_\alpha(Z_{t+1}) = \Phi^{-1}(\alpha),$$

$$\mathrm{ES}_\alpha(Z_{t+1}) = \frac{\varphi\big(\Phi^{-1}(\alpha)\big)}{1-\alpha}.$$

where $\Phi$ and $\varphi$ denote the standard normal CDF and PDF, respectively. On the other hand, one can assume that $Z_{t+1}$ follows a Student-$t$ distribution normalised to have unit variance:

$$Z_{t+1} \overset{d}{=} s_\nu \, T_\nu, \qquad T_\nu \sim t_\nu(0,1),$$

where the normalization scale is

$$s_\nu \;=\; \sqrt{\frac{\nu-2}{\nu}} \qquad (\nu > 2),$$

so that $\mathrm{Var}(Z_{t+1}) = 1$, where $t_\nu(\cdot)$ and $f_{t_\nu}(\cdot)$ denote the CDF and PDF of a Student-$t$ distribution with $\nu$ degrees of freedom. Let $t_\nu^{-1}(\alpha)$ denote its $\alpha$-quantile and define $q_0 := t_\nu^{-1}(\alpha)$ to be the $\alpha$-quantile of the standard (unscaled) Student-$t$. Then, the $\alpha$-quantile and ES for the normalised Student-$t$ are,

$$q_\alpha(Z_{t+1}) = s_\nu \, t_\nu^{-1}(\alpha),$$

$$\mathrm{ES}_\alpha(Z_{t+1}) = s_\nu \underbrace{\left[\frac{\nu+q_0^2}{\nu-1}\,\frac{f_{t_\nu}(q_0)}{1-\alpha}\right]}_{\text{ES of the standard } t_\nu},$$

where $f_{t_\nu}$ is the standard $t_\nu$ density and the bracketed term is the closed-form ES for the standard $t_\nu$. The condition $\nu > 2$ is required for the variance to exist, and hence for the normalization $s_\nu$ to be real. Similarly, one may assume that only the upper tail of $Z_{t+1}$, beyond the threshold $u$, follows a GPD with parameters $(\xi, \beta)$. Under this assumption, for any $\alpha \in [\hat{F}(u), 1)$, the tail probability relative to the exceedance region is

$$p_{\mathrm{tail}}(\alpha) \;=\; \frac{1-\alpha}{1-\hat{F}(u)}.$$

where $\hat{F}(u)$ is the empirical distribution function at the threshold $u$ defined by,

$$\hat{F}(u) \;=\; \frac{1}{N}\sum_{t=1}^{N} \mathbf{1}\{Z_t \le u\} \;=\; 1 - \frac{N_u}{N},$$

where $N$ is the sample size of the training residuals and $N_u = \#\{Z_t > u\}$ is the number of exceedances above the threshold. The $\alpha$-quantile of $Z_{t+1}$ and the corresponding expected shortfall at level $\alpha$ are thus given by:

$$q_\alpha(Z_{t+1}) = u + \frac{\beta}{\xi}\left(p_{\mathrm{tail}}(\alpha)^{-\xi} - 1\right),$$

$$ES_\alpha(Z_{t+1}) = \frac{q_\alpha(Z_{t+1})}{1-\xi} + \frac{\beta - \xi u}{1-\xi}.$$

where $\alpha \geq \hat{F}(u)$ is a necessary condition.

**Note:** *All the quantile and expected shortfall expressions discussed above coincide with those obtained from the fitted parameters.*

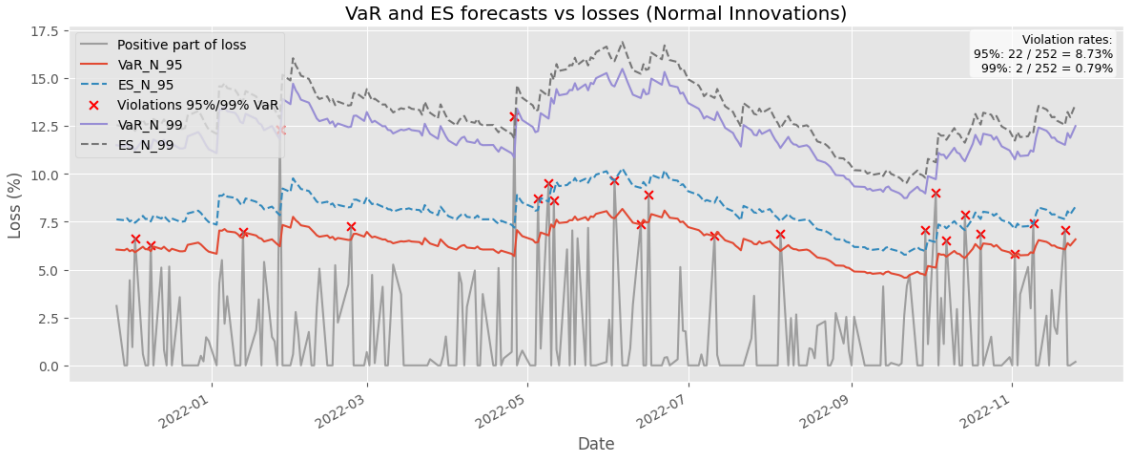



Figure 5: Losses with VaR and ES forecasts at 95% and 99% levels assuming $Z_t \sim N(0,1)$. [Appendix 5]

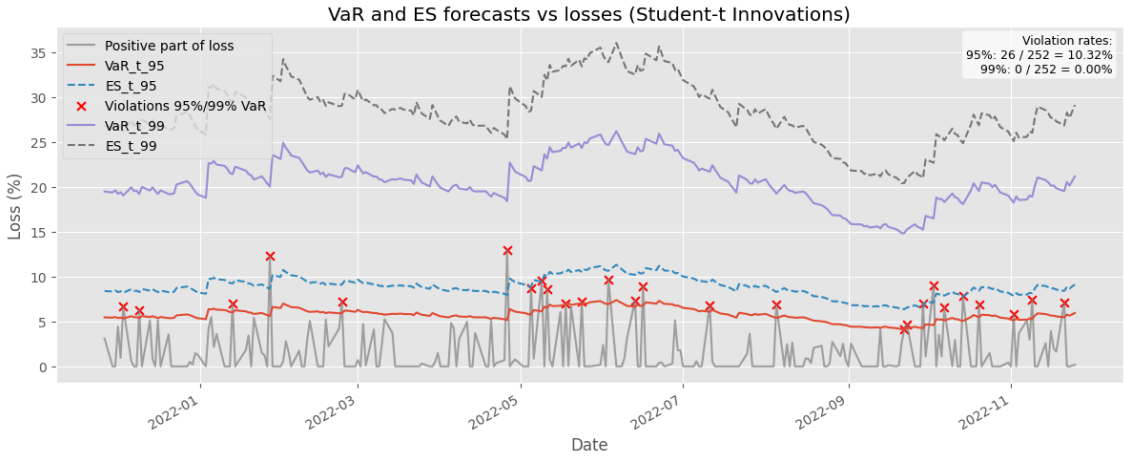



Figure 6: Losses with VaR and ES forecasts at 95% and 99% levels assuming $Z_t \sim t_v$. [Appendix 5]

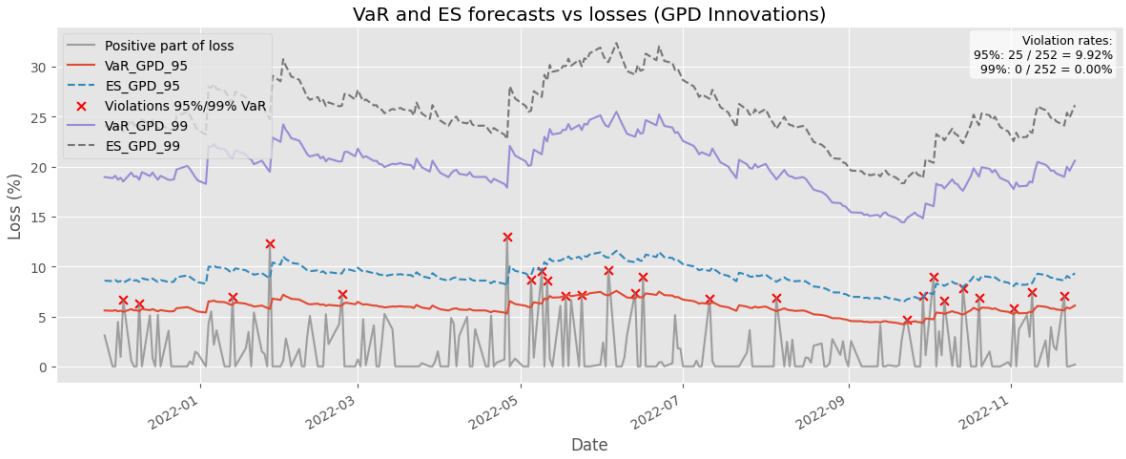



Figure 7: Losses with VaR and ES forecasts at 95% and 99% levels assuming the upper tail of $Z_{t+1}$, beyond the threshold $u$, follows $GPD(\xi, \beta)$. [Appendix 5]

Figures 5, 6 and 7 display the positive losses across the testing data, overlaid with the VaR and ES at the 95% and 99% levels. The violations are marked with red crosses on each graph. The violation rates at each level are given in the top right of each plot. We observe that the VaR and ES for the normal innovation model is much tighter than the other two, which are both comparatively conservative, exhibiting no violations at the 99% level.

# 5 (v) Backtesting

In this section, we present the results of the backtesting for each VaR forecast from the previous section. Backtesting is performed to assess the accuracy of our forecasts. The test we perform is called Kupiec's unconditional coverage test, which evaluates whether the observed frequency of VaR violations aligns with the confidence level ($\alpha$). The null hypothesis for Kupiec's unconditional coverage test is stated as follows:

$$H_0 : E[I_t \mid \mathcal{F}_{t-1}] = 1 - \alpha, \quad t = 1, \ldots, T$$

where $I_t := \mathbf{1}_{\{L_t > \widehat{\mathrm{VaR}}_\alpha(L_t)\}}$, $\widehat{\mathrm{VaR}}_\alpha(L_t)$ is the forecast for the corresponding loss $L_t$, and $\mathcal{F}_{t-1}$ is the $\sigma$-algebra generated by $I_1, \ldots, I_{t-1}$.

| $\alpha$ | Violations | Expected | $\mathrm{LR}_{\mathrm{uc}}$ | $p$-value |
|---|---|---|---|---|
| 0.950 | 22 | 12.6 | 6.097 | 0.014 |
| 0.990 | 2 | 2.52 | 0.117 | 0.733 |
| 0.995 | 2 | 1.26 | 0.370 | 0.543 |
| 0.999 | 2 | 0.252 | 4.802 | 0.028 |

Table 2: Backtesting results for VaR forecast methods assuming $Z_t \sim N(0, 1)$. [Appendix 6]

Tables 2–4 present the Kupiec unconditional coverage test results across multiple confidence levels for VaR models under the three different innovation assumptions. For the normal-innovation case, the 95% level shows a substantially higher number of violations than expected (22 versus 12.6), leading to a significant $LR_{uc}$ statistic and a low p-value (0.014), thereby giving enough

| $\alpha$ | Violations | Expected | $\text{LR}_{\text{uc}}$ | $p$-value |
|---|---|---|---|---|
| 0.950 | 26 | 12.6 | 11.633 | 0.001 |
| 0.990 | 0 | 2.52 | 5.065 | 0.024 |
| 0.995 | 0 | 1.26 | 2.526 | 0.112 |
| 0.999 | 0 | 0.252 | 0.504 | 0.478 |

Table 3: Backtesting results for VaR forecast methods assuming $Z_t \sim t_v$. [Appendix 6]

| $\alpha$ | Violations | Expected | $\text{LR}_{\text{uc}}$ | $p$-value |
|---|---|---|---|---|
| 0.950 | 25 | 12.6 | 10.113 | 0.001 |
| 0.990 | 0 | 2.52 | 5.065 | 0.024 |
| 0.995 | 0 | 1.26 | 2.526 | 0.112 |
| 0.999 | 0 | 0.252 | 0.504 | 0.478 |

Table 4: Backtesting results for VaR forecast methods assuming the upper tail of $Z_{t+1}$, beyond the threshold $u$, follows $GPD(\xi, \beta)$. [Appendix 6]

evidence of incorrect coverage. In contrast, the 99% and 99.5% levels display violation counts close to the expected values, with p-values above the 1% and 5% thresholds. However, at the 99.9% level, the model again underperforms, producing more violations than expected. When assuming Student-t innovations, the 95% and 99% VaR levels perform notably worse. At 95%, violations are more than double the expected count, yielding a highly significant $LR_{uc}$ statistic (p = 0.001). At the 99% level, the complete absence of violations reflects overly conservative forecasts, which also leads to rejection of the null hypothesis. Only the 99.5% and 99.9% levels maintain p-values above the significance thresholds, suggesting reasonably accurate coverage in the extreme tail. The GPD-based model follows a similar pattern to the Student-t model. At the 95% level, the number of violations is again roughly double the expected value. At the 99% level, the lack of any violations produces a significant $LR_{uc}$ value. Much like the Student-t innovations, the 99.5% and 99.9% confidence levels show enough evidence of satisfactory tail performance when modelling exceedances with the GPD.

These results align with theoretical expectations of the model assumptions. The light-tailed normal distribution tends to underestimate tail risk at higher confidence levels ($\alpha \geq 0.99$), whereas heavy-tailed distributions such as the Student-t and GPD produce more conservative forecasts. For moderate thresholds ($\alpha < 0.99$), the inverse holds: the normal model overestimates coverage, while Student-t and GPD models tend to overpredict risk. The similarity in performance between the two heavy-tailed specifications is consistent with the estimated tail index $\widehat{\xi} = 0.138$, indicating that the data exhibits only moderate tail heaviness. Consequently, the Student-t distribution may suffice for modeling purposes, as it captures the heavy-tailed nature of the returns while effectively describing the body of the data. Nevertheless, the GPD framework remains valuable due to its theoretical robustness, asymptotic justification for modeling extreme exceedances, and potential resilience under more volatile future market regimes.

We conclude with a remark that this coursework provides Bob with a stable, parametric, and computationally efficient tool for measuring and forecasting risk backed by theoretical asymptotic analysis of extreme behaviour. This draws a stark contrast with our previous coursework where we focused on empirical aspects with heuristics as we recomputed models on a 500-day rolling window based on filtered simulations. The methodology presented in this work is suited for regulatory reporting purposes because it allows us to evaluate extreme behaviour over longer time horizons, at the cost of immediate adaptation to volatility. On the other hand, our previous work for Bob

may be more suited for a trading operation or a risk desk that requires constant internal monitoring of current market conditions.

## Acknowledgements

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